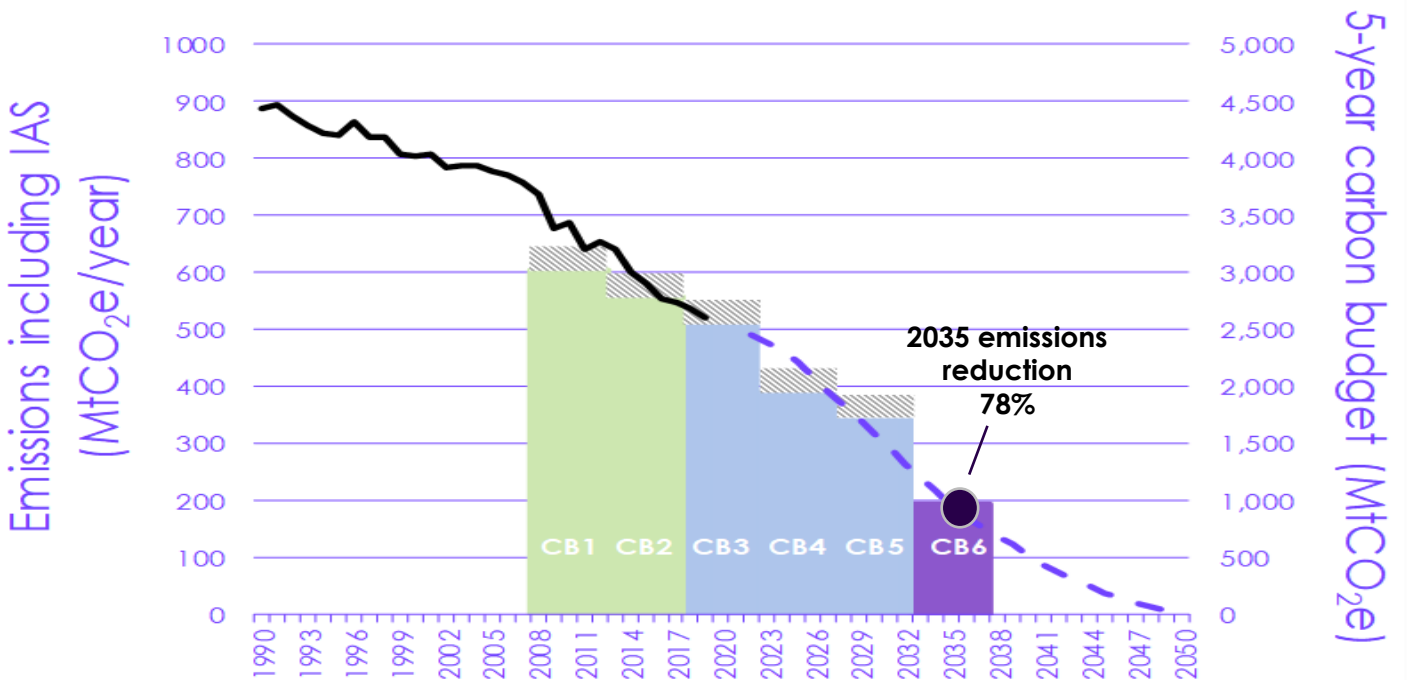
Farm Woodland Forum
8th June 2021

Role of agroforestry and hedges in UK climate targets

Indra Thillainathan
Climate Change Committee

Our recommended path for achieving Net Zero GHG emissions in the UK by 2050

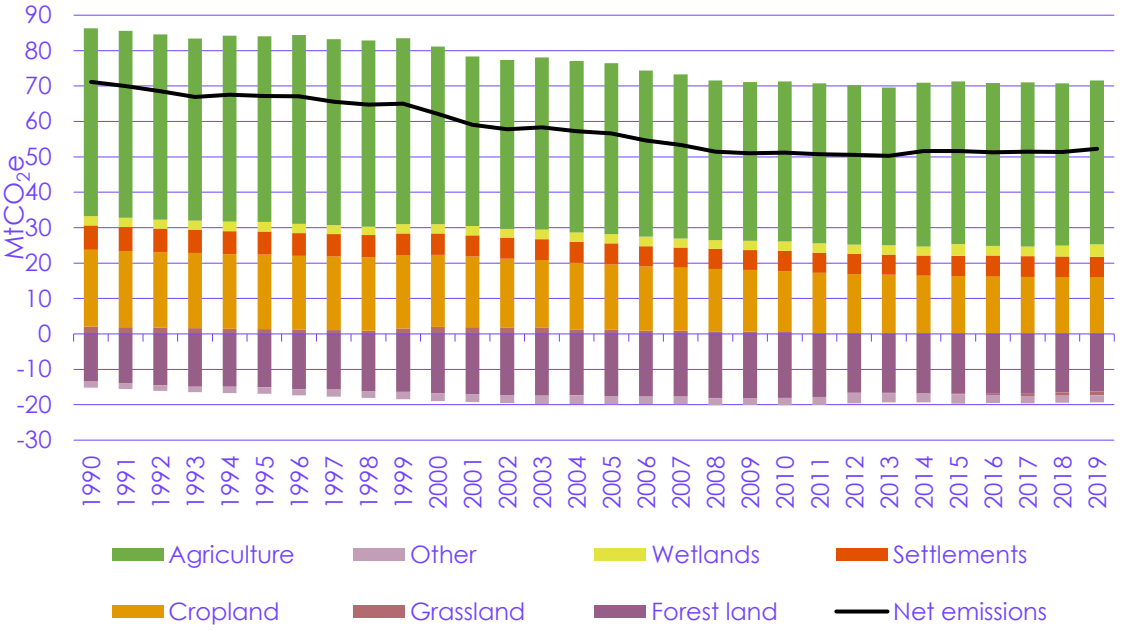
- The UK has a legislated target to reduce GHG emissions to Net Zero by 2050
- In December, we provided advice on the volume of emissions the UK can emit during the period 2033-2037 (Sixth Carbon Budget).

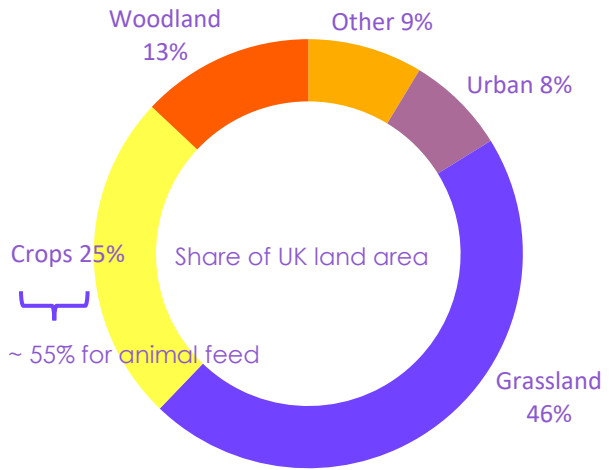


Source: CCC analysis.

Agriculture and land use account for ~ 11% of UK GHG emissions in 2019

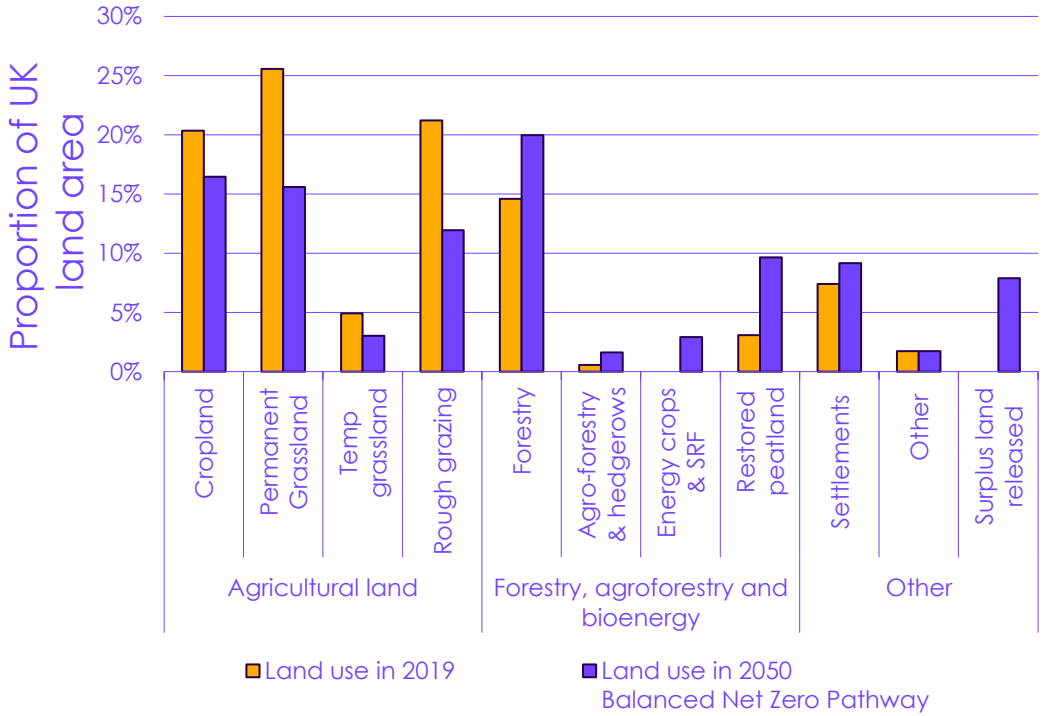
Over 70% of UK land area is used for farming





Changes in land use
Transformation in land use now to 2050

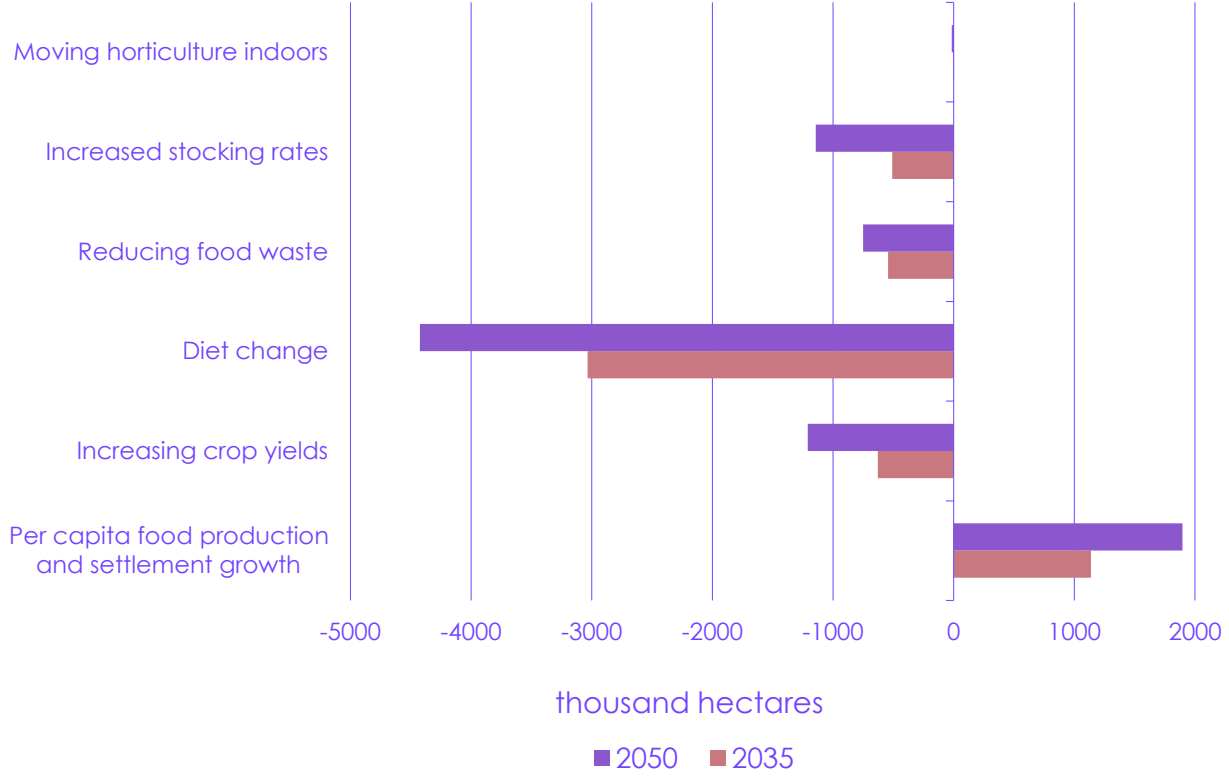
- Around one-third of agricultural land is freed up through changes in output and more efficient farming practices.
- We need 21% of this land for actions to sequester and reduce CO_2.
- In total, 25% of the UK land area is forested or used for agro-forestry and energy crop production by 2050 - compared to around 15% today.
- Harmful peat extraction is ended, and nearly 80% of the UK's peatlands are restored.



Source: CCC Analysis

Agricultural land released in the Balanced Pathway

3.8 million hectares (21%) by 2035 and 6 million hectares (35%) by 2050.



Transformation of land use is needed - requires the release of a fifth of agricultural land by 2050 for these other measures

Planting trees…



50,000ha/year by 2035

…and on-farm (agroforestry) & more hedges



Trees on 10% of farm land, extend hedges by 56kha by 2050

Energy crops



30,000/year by 2035

Restoring peat



80% restored by 2050

Agroforestry on 10% of farmland by 2050
Sixth Carbon Budget assumptions

- The low planting densities of agroforestry systems results in 14% of the grassland area and 7% of cropland area dedicated to these systems.
- **Silvoarable** systems plant poplar YC12 in two-metre-wide rows, a spacing of 30 metres between each row and seven metres between each tree. The spacing takes account of the need to minimise shading which can adversely impact crop yields.
- **Silvopastoral** systems are planted with broadleaf species (e.g. sycamore, ash and birch) with a YC6, and at a higher planting density of 400 trees per hectare.

Results:

- **Annual carbon sequestration rates of 1 $MtCO_2e$ in 2035, rising to 2.3 MtCO2e in 2050.**



Hedgerows
Sixth Carbon Budget assumptions

- Current hedgerow length is 120,000 hectares (over half is managed)
- Create an additional 28,000 hectares by 2035 and 56,000 hectares by 2050.*
- 30% of the new length is managed for biomass fuel.
- All hedge creation is on grassland.

Results: Annual carbon sequestration rates of 0.3 $MtCO_2e$ in 2035(assumes no change in soil carbon)

Wider environmental benefits

- Biodiversity: providing food and shelter for a range of birds, insects and mammals.
- Facilitate movement through the landscape – connectivity.
- Regulating services:
  - Air quality
  - Water quality
  - Reduce flood risk
  - Reduce soil erosion
  - Natural pest control and polinators

Note: *Hedges are assumed to be 1.5 m wide and with biomass stock densities derived from the BEIS Biomass Extension project (2014).

The Balanced Pathway in Land use

Net emissions reduce by 93% to around 1 $MtCO_2e$ by 2035 and is a net sink (19 $MtCO_2e$) by 2050

Agroforestry and hedges account for around 10% of annual abatement

